Candidate Seat No: _____

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
CONTINUOUS INTERNAL PRACTICAL EVALUATION – 1
FIFTH SEMESTER OF B.TECH. (IT)
DATA ENGINEERING [ITUC301]

Date: 24/08/2026, Monday **Time: 20:20 PM to 4:20 PM** **Maximum Marks: 10**

Problem Definition

An e-commerce organization collects data from multiple sources, including:

- Customer and product information stored in a relational database.
- Transaction data received through an API in JSON format.
- Web application user activity logs generated continuously.

You are required to design and implement a simplified end-to-end data engineering pipeline.

Tasks to be Performed

1. **Data Engineering Lifecycle Design – 2 Marks**
   Draw or explain the flow of data through the following stages:

Data Generation → Data Ingestion → Storage → Transformation → Data Serving

Also identify one consideration related to data quality, security, privacy, or observability in the pipeline.

2. **Data Generation and Profiling – 3 Marks**
   Using Python, create or use sample data from at least two different data sources, such as:
   - CSV/relational data, and
   - JSON/API/log data.

Perform basic data profiling and identify data quality issues such as:

- Missing/null values
- Duplicate records
- Invalid or inconsistent data formats

3. **Batch and Streaming Ingestion – 4 Marks**
   Implement:
   - A batch ingestion process to read and process data periodically, and
   - A streaming-style ingestion process that processes incoming events one at a time or continuously.

Compare the two approaches based on:

- Data processing latency
- Throughput
- Suitable use cases

4. **Result and Analysis – 1 Mark**
   Provide a brief conclusion explaining which ingestion approach would be more suitable for:
   - Historical transaction processing, and
   - Real-time user activity monitoring.